MINOR PROJECT
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AI SMART KNOWLEDGE MANAGEMENT PLATFORM

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Name	**Shah Prem JigneshKumar** **Shah Jeet MehulBhai** **Shah Jash PriteshKumar**
Branch	**B.Tech (C.E)**
Guide Name	**Prof. Kruti Sheth**

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Introduction

AI SMART KNOWLEDGE MANAGEMENT PLATFORM

Problem: Knowledge is scattered across PDFs, research papers, videos, documents, notes, images, spreadsheets, and web content, making it difficult to find, organize, and understand information. Solution: An AI-powered Knowledge Workspace that stores all your knowledge, understands uploaded content, provides AI-powered search, generates summaries, quizzes, and flashcards, and supports interactive conversations with documents

Background and Motivation

Background

Current challenges: large unorganized information volume , repetitive document searching , and time – consuming manual note creation . No unified AI workspace exists , and existing tools offer limited document understanding.

Motivation

Build a Personal AI Research Assistant , AI Knowledge Manager , Learning Companion , and Intelligent Search Engine — inspired by Google AI Studio , Note book LM , Chat GPT, Perplexity AI , and Microsoft Copilot.

Relevance and Importance

Relevance

Provides a centralized AI-powered platform to manage and understand diverse digital information. Helps users search, summarize, organize, and interact with their documents and knowledge efficiently. Combines AI to provide relevant and source-based answers.Useful for students, researchers, professionals, and organizations for improving productivity and learning.

Importance

AI-powered knowledge management is reshaping how individuals and organizations interact with information. The growing demand for intelligent productivity tools reflects a need for systems that not only store content but actively understand and assist users. This platform addresses that demand by unifying search, learning, and document intelligence in a single workspace.

Literature Review

- Google AI Studio provides a platform to interact with AI models, test prompts, and work with different types of inputs.
- NotebookLM allows users to upload documents and ask questions, generate summaries, quizzes, and study materials from their sources.
- Google Gemini provides text and image understanding, question answering, summarization, and content generation.
- ChatGPT provides conversational interaction and helps users understand and generate content.
- Traditional document systems mainly focus on storing and organizing files, with limited interactive learning features.
- However, the set tools combine knowledge management, document intelligence, AI search , long term memory, and multi format understanding into a single unified work space.

Literature Review

Rank	Ref. No.	Paper	Api/Technology	Why It Best Suits Your Project	How You Can Use IT(Findings)	Refrences papers link
1	**https://ray.yorksj.ac.uk/id/eprint/9863/1/06.pdf**	Domain Pro Chatbot: A Multi-Domain AI Career Advisor System	Google Gemini API, Python, Flask, SQLite	Strongest practical match: a lightweight, self-hosted chatbot using Gemini, user accounts, stored conversations, and minimal infrastructure.	Use as the main implementation reference for Gemini API integration, authentication, chat storage, and a web-based chatbot architecture.	https://www.ijirset.com/upload/2024/april/33_Personalized.pdf
2	**https://journals.ekb.eg/**	Implementation of the Gemini LLM in Android-Based Chatbot Application Development	Gemini API key, Android	Directly studies implementation of a Gemini API key in a chatbot and verifies whether the application can use Gemini functionality successfully.	Cite it to justify that your system can rely on a hosted Gemini model rather than training an LLM. Adapt the Android implementation concept to your web application.	https://e-journal.gomit.id/jsseis/article/view/70
3	**https://zuscholars.zu.ac.ae/cgi/viewcontent.cgi?article=6907&context=works**	Personalized Chatbot Optimization Using OpenAI	OpenAI API, user-oriented chatbot personalization	Direct match for your personalization goal: an application-level chatbot using an external API. The exact published title is Personalized Chatbot Optimization Using Open AI developer android	Use in the literature review to show that personalization can be handled through user context and prompt design, without claiming RAG.	https://ijcrt.org/viewpaperforall.php?paper=IJCRT2609024
4	**Developer android**	AI Chatbot Using the MERN Stack and OpenAI API	MongoDB, Express.js, React, Node.js, OpenAI API	Relevant full-stack example: web interface, backend server, database, and LLM API communication. Related implementations use the MERN stack with OpenAI to create full-stack chatbot applications.	Use it for your system-design chapter: frontend â†' backend â†' LLM API â†' generated response â†' chat UI.	https://ijrpr.com/uploads/V6ISSUE7/IJRPR50882.pdf
5	**Developer android**	Voice Command Chatbot with ChatGPT Integration	Python, Flask, OpenAI API, Gemini API, Google TTS / pyttsx3	Shows a chatbot that integrates external LLM APIs through a backend service; the paper lists Python, Flask, OpenAI API, Gemini API, and text-to-speech technologies.	Use it to explain secure server-side API calling, prompt construction, and future work such as voice input/output. Do not list voice as a current feature unless you have implemented it.	https://www.ijsdr.org/papers/IJSDR2511109.pdf

Rank	Ref. No.	Paper	Api/Technology	Why It Best Suits Your Project	How You Can Use IT(Findings)	Research paper Link
6	**Developer android**	Mind Mate: An AI-Powered Chatbot for Mental Health Support	Gemini API, web/mobile chatbot backend	Uses Gemini API to provide user-facing, personalized conversational support.	Use it to support the claim that Gemini can generate context-sensitive and personalized chatbot responses. Clearly state that your project is for knowledge management, not medical or mental-health advice.	https://ijrpr.com/uploads/V5ISSUE5/IJRPR27476.pdf
7	**google**	GENIE: Intelligent Chatbot	MERN stack, OpenAI API, GPT-based generation	A web application built using MERN that integrates OpenAI APIs for text and code generation.	Use it as an additional full-stack reference for responsive UI, API routing, environment-variable management, and calling an external LLM from a backend.	https://www.ijcrt.org/papers/IJCRT24A5643.pdf
8	**Developer android**	Selection of Large Language Model for Development of Interactive Chat Bot for SaaS Solutions	ChatGPT/OpenAI API, Cohere API, Llama API comparison	Useful for explaining why a project uses a hosted LLM API instead of building a model from scratch. The study evaluates three API-accessible models and recommends ChatGPT based on speed, cost-effectiveness, and relevance. Developer android	Use in the methodology or technology-selection section to compare Gemini, OpenAI, Cohere, and other LLM APIs by cost, response quality, availability, and ease of integration.	1890+Jurnal+IJSET+Lianawati+Suwito.pdf
9	**journal.umgo.ac**	Cooking and Food Information Chatbot System Using GPT-3	GPT-3, OpenAI API	A domain-specific chatbot that uses GPT-3 to provide food-related information and personalized recommendations. The Cook Pal chatbot was developed to support healthy-eating information and dietary concerns. Developer android	Use it to show that a general-purpose LLM API can be adapted to a specialized domain by using structured prompts and user inputs. Your specialized domain is personal and organizational knowledge access.	https://rjwave.org/jaafr/papers/JAAFR26A5269.pdf
10	**journal.eng.unila.ac**	Thinking Assistants: LLM-Based Conversational Assistants	GPT-4, OpenAI Streaming API	Relevant for responsive multi-turn conversation, interactive assistant behavior, and user-facing chat design.	Use it when discussing chat history, conversation flow, streaming responses, usability, and future enhancement of the chatbot interface.	https://www.ijisrt.com/assets/upload/files/IJISRT25SEP264.pdf

Research Gap

Previous studies have shown that external Large Language Model APIs, including Gemini and OpenAI APIs, can be integrated into web and mobile applications to provide interactive, contextual, and personalized chatbot responses. Existing systems commonly focus on individual domains such as customer service, career guidance, mental-health support, website assistance, or library services. However, many do not combine API-based personalized chat interaction with a general-purpose knowledge management environment that supports secure user access, centralized document organization, categorization, document-update tracking, and unified information access. The proposed Smart Knowledge Management System addresses this integration gap by providing a full-stack platform that combines document management and secure knowledge sharing with an external API-based personalized chatbot. The current system uses API-key integration and prompt-based interaction, without implementing RAG, vector databases, Lang Chain, embeddings, or a custom NLP pipeline.

Objectives

Objectives

Primary objectives focus on building an AI - powered knowledge workspace that supports multiple document formats, implements semantic search, enables AI conversation switch documents, and integrates YouTube knowledge extraction . Secondary objectives include generating summaries, creating flash cards, building quizzes, maintaining long term workspace memory, and providing source AI-responses .

Required Tools and Technology

Required Tools and Technology

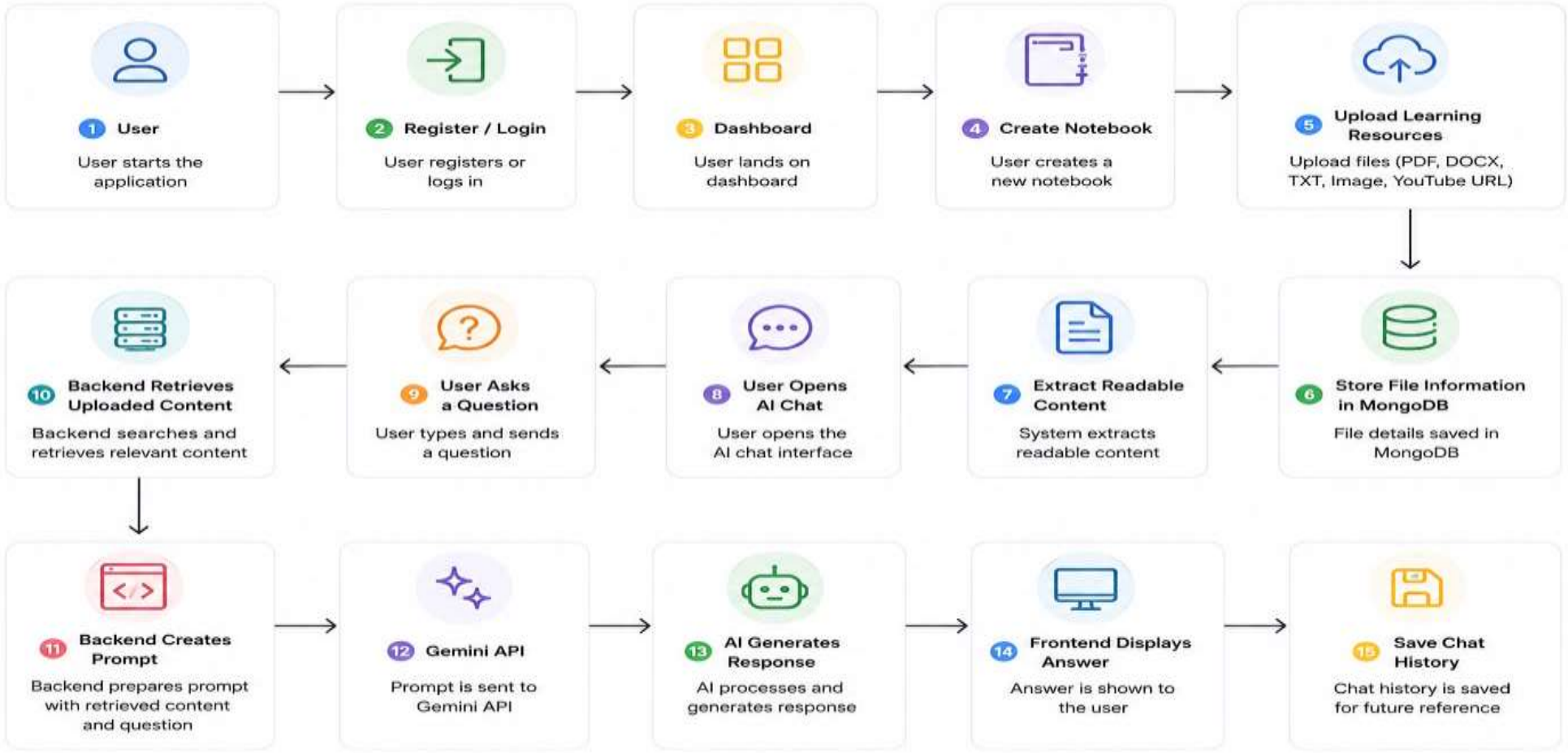
Frontend	Backend	Database	AI Integration
Next.js 15	FastAPI	MongoDB	Google Gemini API
React.js	Python		Google Generative AI SDK
TypeScript	Uvicorn		
Tailwind CSS	Pydantic		
shadcn/ui	JWT Authentication		

Method/ Approach

Methodology

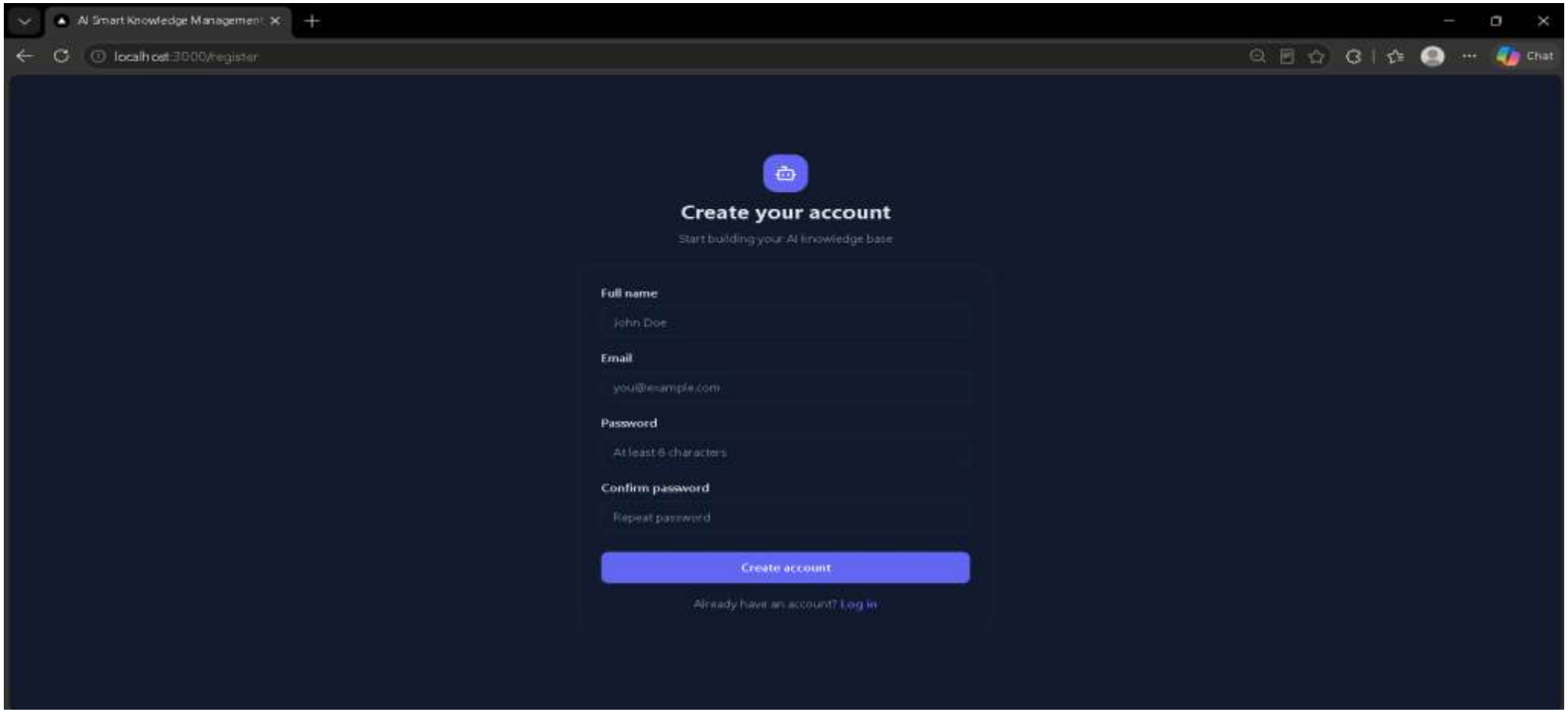
1.Upload document or video
2.Content extraction from files
3.Text processing and cleaning
4.Embedding generation for semantic representation
5.Store embeddings in vector database
6.Semantic indexing for retrieval
7.AI retrieval using Retrieval Augmented Generation (RAG)
8.Response generation with accurate

System Architecture



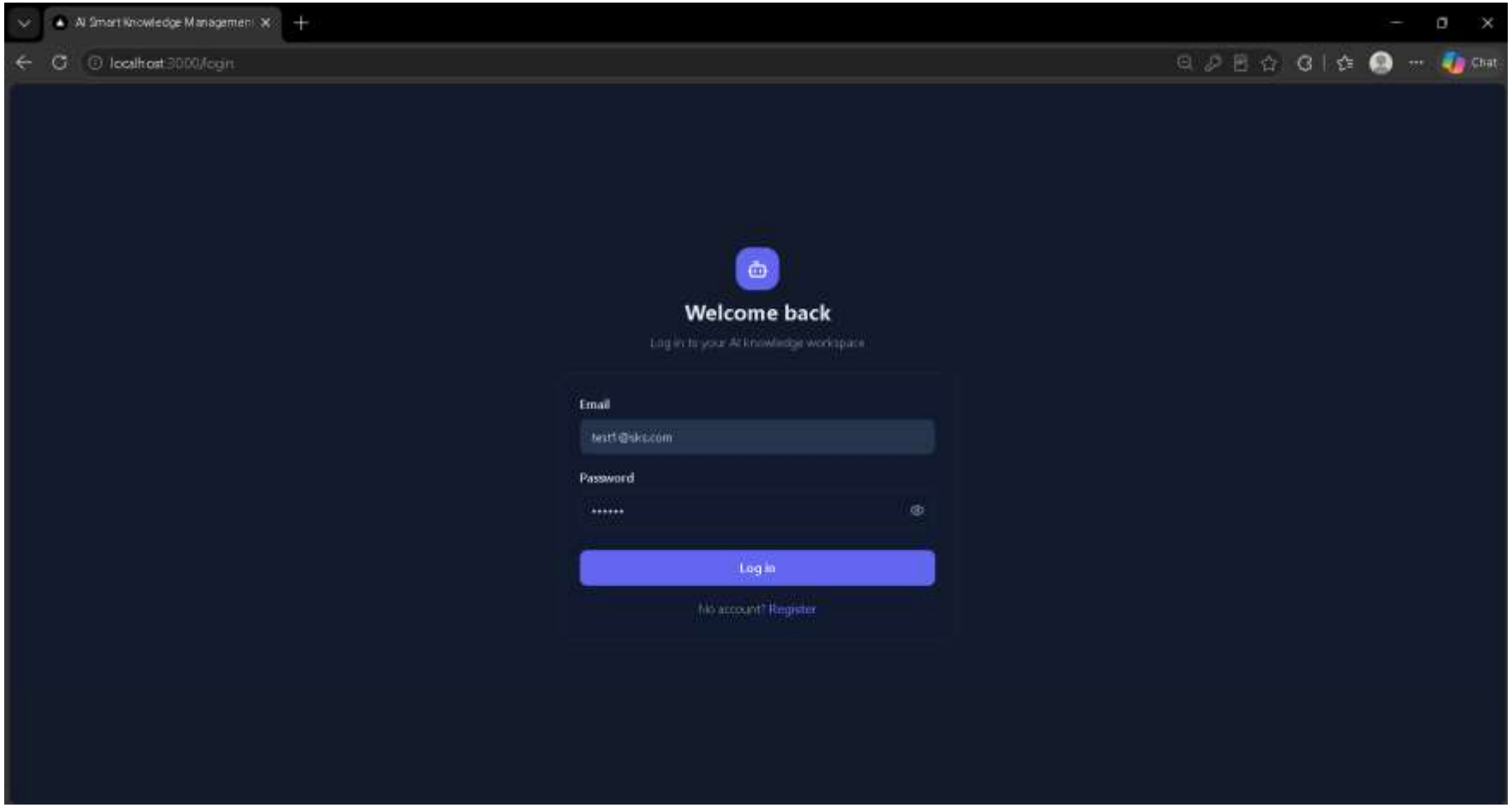
Result

Register page



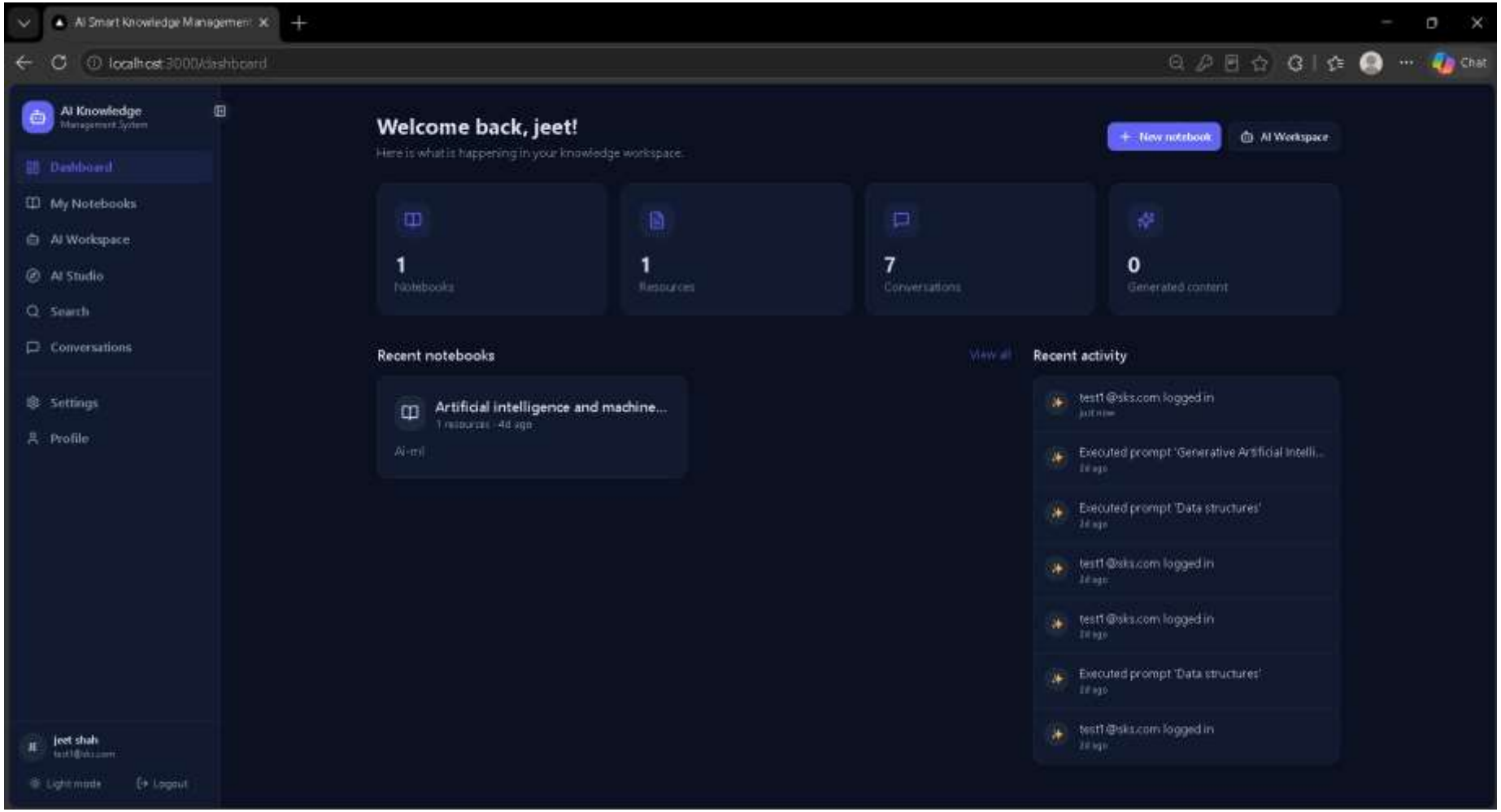
Result

Login page



Result

Dashboard



Future work /Scope

- Add document version tracking so users can compare updatedresources.
- Add resource-level permissions for shared organizationalnotebooks.
- Introduce richer source references in the chat interface so userscan quickly inspect the original material.
- Add multilingual interaction for regional and internationaleducational use.
- Provide export of summaries, quizzes, and flashcards intoprintable study formats.
- Add analytics showing which notebooks and learning activitiesare used most often.
- Develop automated regression tests for resource processing andchat workflows.

Conclusion

This project introduces a Smart Knowledge Management System that centralizes scattered learning materials into a single interactive platform. By smoothly integrating the Gemini API and database storage into a practical workflow, it helps users actively learn from their resources rather than just storing them. As a straightforward academic prototype, it proves that AI works best when embedded directly into everyday tasks. Moving forward, future work will focus on real-world user testing, stronger privacy controls, better resource processing, and clearer source tracking to fully measure and expand the system's impact.

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Refrences

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Thank you